

Continuous Convergence and Curried Functional Spaces

1. Preliminary definitions

Definition 1.1. Let X be a set. A *filter* on X is a non-empty set $F \subset 2^X$ which is closed under supersets and finite intersections, and does not contain the empty set. The set of all filters on X will be denoted by $\mathcal{F}(X)$.

Definition 1.2. Let X be a set. A family $\mathcal{B} \subset 2^X$ is called a *filter base* if it does not contain the empty set and is closed under finite intersection. The filter $[\mathcal{B}]$ generated by $\mathcal{B}$ is defined as the collection of all supersets of sets from $\mathcal{B}$.

If $\mathcal{B} = \{\{x\}\}$ where $x \in X$, the filter $[x] = [\{\{x\}\}]$ is called the *universal filter* of x .

If $f : X \rightarrow Y$ is a map and $F \in \mathcal{F}(X)$ is a filter on X , then $\{f(A) \mid A \in F\}$ is a filter base that generates the *image filter* denoted $f(F)$.

Definition 1.3. Let X be a topological space. Then we say that a filter F on X *converges* to a point $x \in X$, and write $F \rightarrow x$, if it contains all open neighborhoods of x .

Remark 1.1. The language of filters is expressive enough to encode all topological properties. For example [1]:

- A sequence $\{x_n\}_{n=1}^{\infty}$ converges to a point $x \in X$ iff the filter

$$F = \{A \subset X \mid x_n \in A \text{ for all but finitely many } n\}$$

converges to x . This filter F is called the *derived filter* of $\{x_n\}_{n=1}^{\infty}$.

- A subset $E \subset X$ is closed if for every point $x \in E$ there is a filter $F \in \mathcal{F}(X)$ converging to x such that $A \cap E \neq \emptyset$ for all $A \in F$.
- A topological space X is compact iff every filter F on X is contained in a convergent filter on X .
- A function $f : X \rightarrow Y$ between topological spaces is continuous iff it preserves convergent filters, i.e. $F \rightarrow x \in X$ implies $f(F) \rightarrow f(x) \in Y$.

Definition 1.4. One may try to decouple the notion of convergent filters from topology. Let X be a set with a relation $\rightarrow \subset \mathcal{F}(X) \times X$. Then $(X, \rightarrow)$ is called a *convergence space* if

- For all $x \in X$, we have $[x] \rightarrow x$;
- If $F \rightarrow x$ and $G \rightarrow x$, then $F \cap G \rightarrow x$;
- If $F \rightarrow x$ and $F \subset G$, then $G \rightarrow x$.

Definition 1.5. A map $f : X \rightarrow Y$ between convergence spaces is said to be *continuous* if $f(F) \rightarrow f(x) \in Y$ whenever $F \rightarrow x \in X$.

Definition 1.6. Let X, Y be two convergence spaces. Then the product $X \times Y$ can be endowed with the following convergence structure: a filter $F \in \mathcal{F}(X \times Y)$ converges to $(x, y) \in X \times Y$ if and only if $p_X(F) \rightarrow x \in X$ and $p_Y(F) \rightarrow y \in Y$.

Definition 1.7. Let X, Y be convergence spaces. By $\mathcal{C}(X, Y)$ denote the set of all continuous functions from X to Y . Then $\mathcal{C}(X, Y)$ can be endowed with a convergence structure as follows. For a filter $F \in \mathcal{F}(\mathcal{C}(X, Y))$, we say that $F \rightarrow f \in \mathcal{C}(X, Y)$ if and only if $F(P) \rightarrow f(p)$ whenever $P \rightarrow p \in X$. Here, the filter $F(P)$ is based on

$$\{A(B) \mid A \in F, B \in P\} = \{\{a(b) \mid a \in A, b \in B\} \mid A \in F, B \in P\}.$$

The resulting convergence structure on $\mathcal{C}(X, Y)$ is called the structure of *continuous convergence*.

Remark 1.2. Continuous convergence on $\mathcal{C}(X, Y)$ is the coarsest convergence structure that makes the *evaluation mapping*

$$\omega : \mathcal{C}(X, Y) \times X \rightarrow Y$$

continuous. Here, the product $\mathcal{C}(X, Y) \times X$ is endowed with the product convergence structure.

Proposition 1.1. (see [2], page 26): *Let X, Y, Z be convergence spaces. Then a map $h : X \rightarrow \mathcal{C}(Y, Z)$ is continuous if and only if the associated map*

$$\tilde{h} : X \times Y \rightarrow Z$$

defined by $\tilde{h}(x, y) = h(x)(y)$, is continuous.

Proof: Assume that h is continuous. To show the continuity of $\tilde{h}$, it suffices to take two convergent filters $F \rightarrow x \in X$ and $G \rightarrow y \in Y$, and show that $\tilde{h}(F \times G) \rightarrow \tilde{h}(x, y)$, where the filter $F \times G$ is based on $\{A \times B \mid A \in F, B \in G\}$. Indeed, we have

$$\tilde{h}(F \times G) = h(F)(G) \rightarrow h(x)(y) = \tilde{h}(x, y),$$

since $h(F) \rightarrow h(x)$ by the continuity of h . The converse is proven similarly. ■

Proposition 1.2. (see [2], page 34): *Let X, Y be topological spaces such that X is locally compact and Y is regular. Then the convergence space $\mathcal{C}(X, Y)$ is topologized by the compact-open topology. That is, the convergence relation arising from this topology coincides precisely with the convergence structure given in [Definition 1.7](#).*

2. Curried functional spaces

Proposition 2.1. *Let X, Y be topological spaces such that X is first countable. Then a sequence $\{f_k\}_{k=1}^\infty \subset \mathcal{C}(X, Y)$ converges to a function $f \in \mathcal{C}(X, Y)$ (in the sense that its derived filter converges) if and only if $f_k(x_k) \xrightarrow[k \rightarrow \infty]{} f(x)$ whenever $x_k \xrightarrow[k \rightarrow \infty]{} x$ in X .*

Proof: First, let F be the derived filter of $\{f_k\}_{k=1}^\infty$ and suppose that F converges to a function $f \in \mathcal{C}(X, Y)$. Consider a sequence $\{x_k\}_{k=1}^\infty$ converging to $x \in X$. ■

Bibliography

- [1] N. Bourbaki, *General Topology, Part I*. 1966.
- [2] R. Beattie and H.-P. Butzmann, *Convergence Structures and Applications to Functional Analysis*. 2002.